

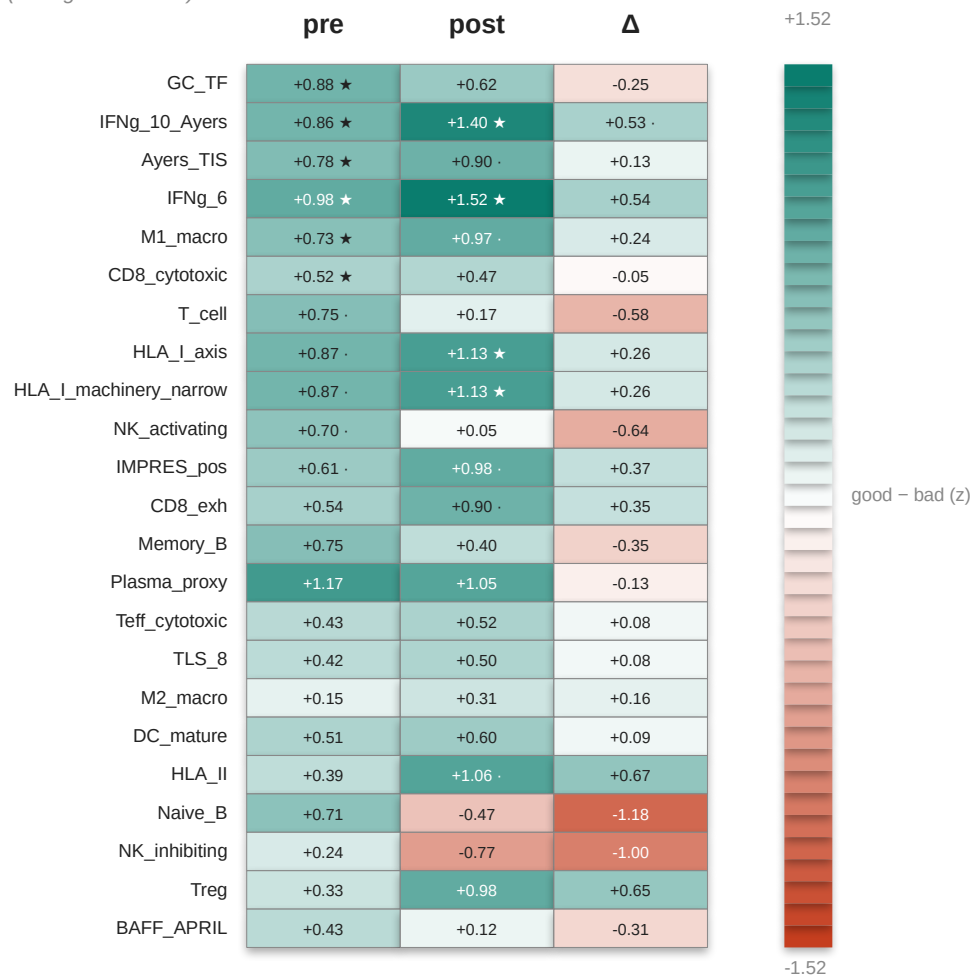
Supp Fig S22. NanoString PanCancer Immune orthogonal validation

6-subject extreme-phenotype (3 pCR vs 3 poor) × paired pre/post × 730 probes.

Six panels: A pre/post/ Δ heatmap · B pre-treatment direction waterfall · C regulatory-grade signatures · D IAE vs IBI composite · E IAE vs IBI genes
· F Pre-spec cascade arrow 4 (Δ CD8-exh × Δ TLS-8, $r=+0.82$, $P=0.046$).

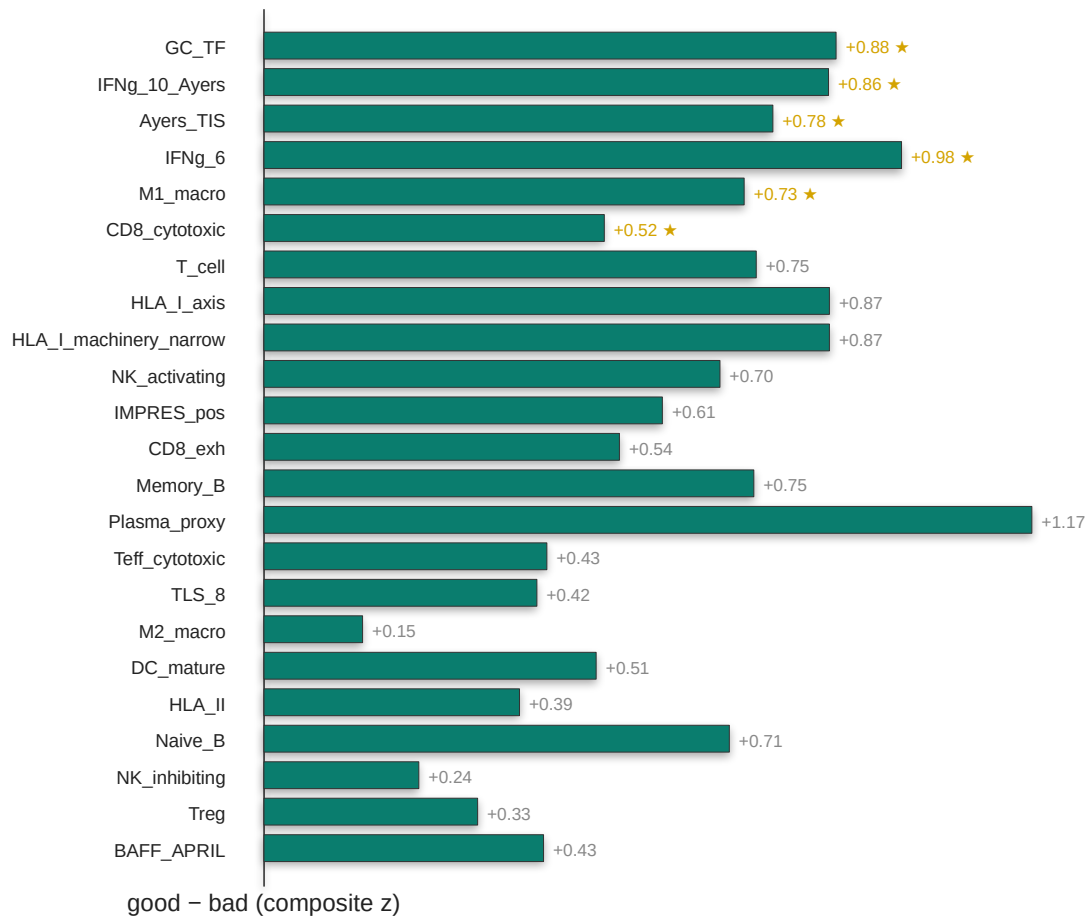
A Pre/post/ Δ composite heatmap (good – bad)

Star ★ = one-sided MW $P \leq 0.05$ (ceiling at $n=3$ vs 3). Dot · = $P \leq 0.10$.



B Pre-treatment direction waterfall — 23/23 good > bad

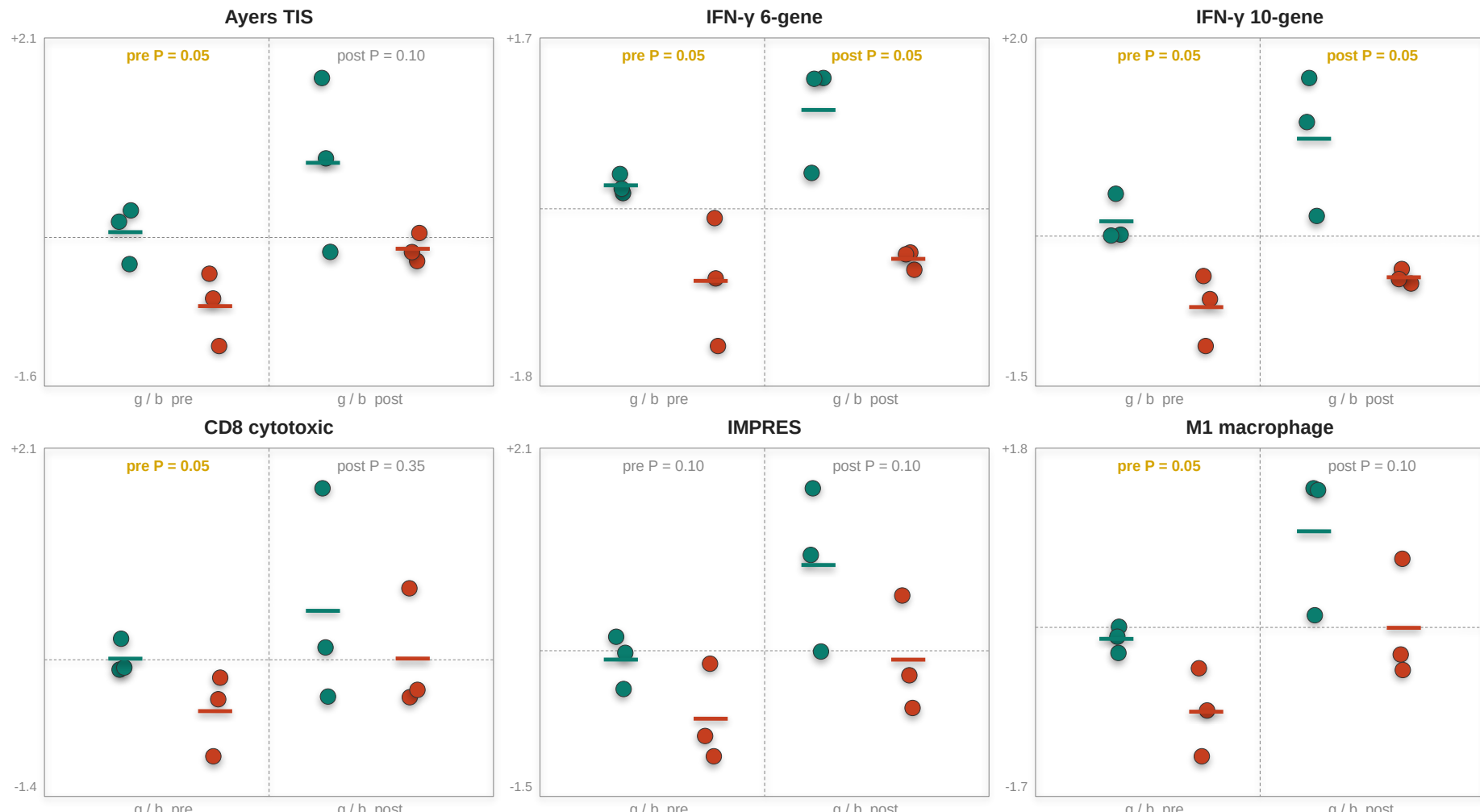
Six composites reach ceiling $P = 0.05$ (Ayers TIS, IFN- γ 6/10, CD8 cytotoxic, M1 macrophage, GC-TF).



★ = one-sided MW $P \leq 0.05$ (ceiling)

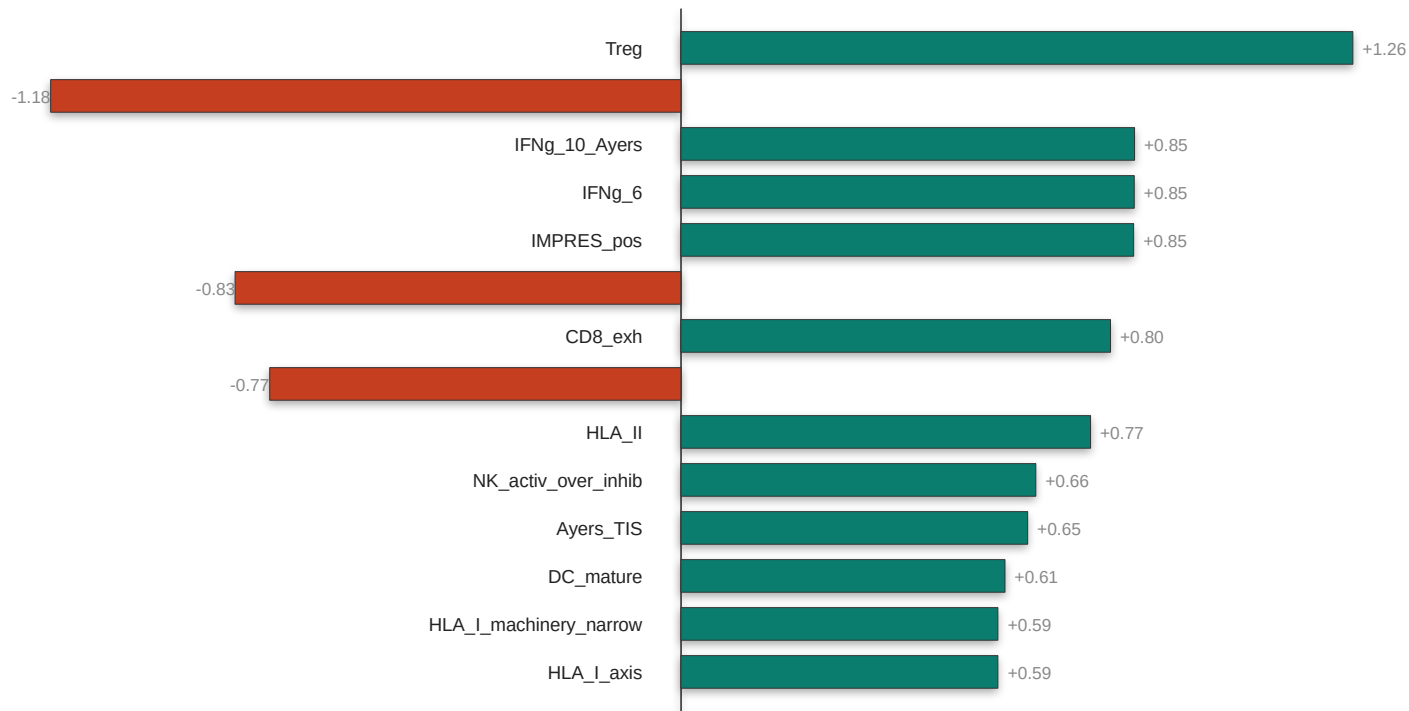
C Regulatory-grade signatures

Six FDA/regulatory-grade immune signatures: pre-RT (left) vs post-RT (right), good (teal) vs bad (coral).



D IAE (n=2) vs IBI (n=3) — composite fingerprint

Positive (teal) = IAE > IBI; negative (coral) = IBI > IAE.



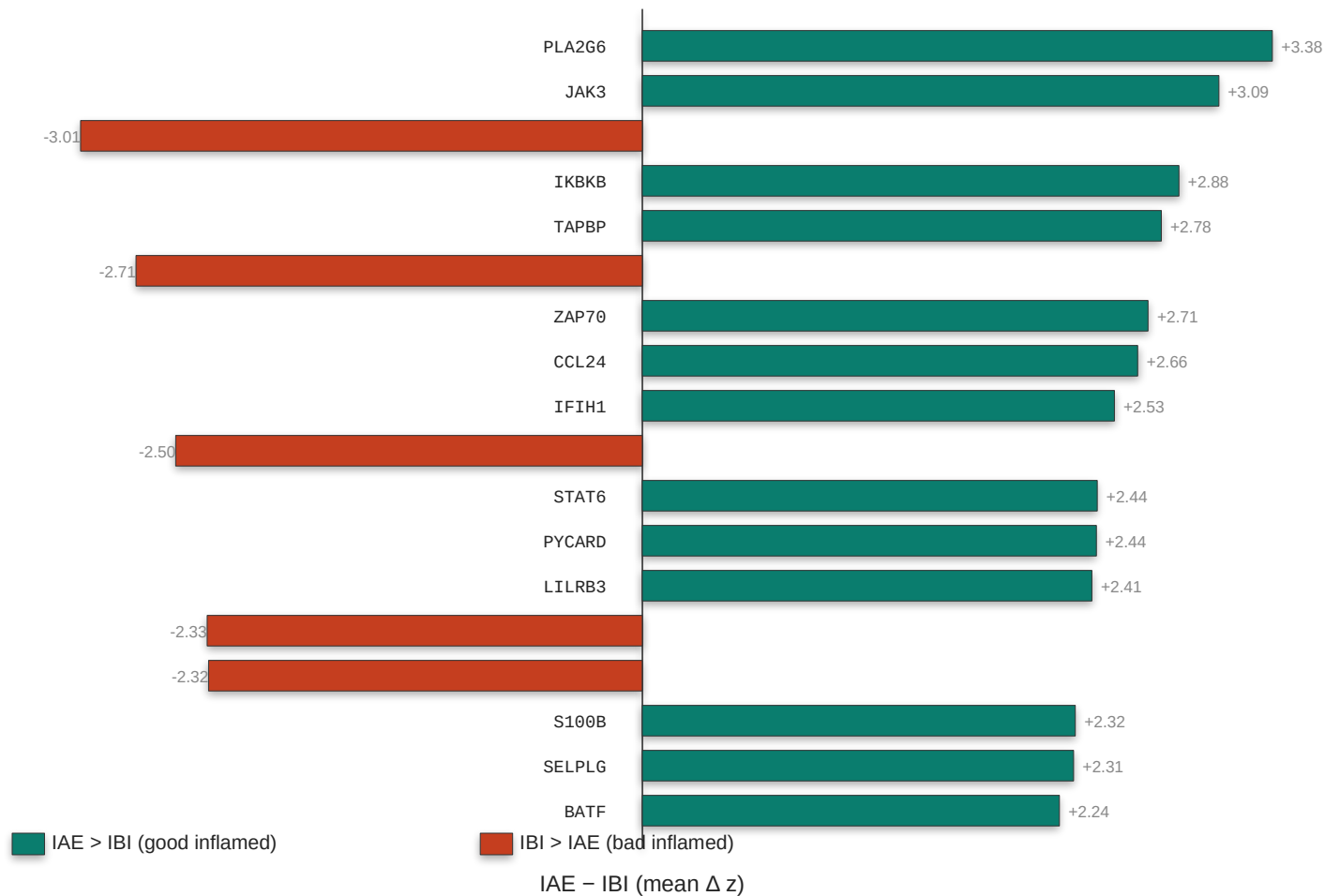
IAE – IBI (mean Δz)

IAE > IBI (good inflamed)

IBI > IAE (bad inflamed)

E Top 18 gene discriminators

IAE productive cascade (JAK3, IKBKB, TAPBP, ZAP70, BATF) vs IBI suppressive (KIR_Inh, ARG2, RAG1).



F Pre-spec cascade arrow 4 (T2): $\Delta\text{CD8-exh} \times \Delta\text{TLS-8}$

Pearson $r = +0.82$, $P = 0.046$ · Spearman $\rho = +0.77$, $P = 0.072$ ($n = 6$ paired)

